Supplementary information for: Thermodynamic modeling tools for the interpretation of melt inclusions and volcanic gases

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1 List of Supplementary Materials

All of these materials are available through the journal's repository and in the GitHub repo: `https://github.com/kaylai/volcanic-gas-modeling-tool-comparison`.

1. This PDF
2. All modeling results from all tools generated in this work: in folder "`Model_Outputs`"
3. All modeling tool settings used to generate results: in folder "`Model_Outputs/resolved_inputs`"
4. Notebooks for creating every figure in this manuscript: in folder "`Figure_Notebooks`"
5. Full datasets used to choose starting compositions for our four basaltic systems: "`Melt_Inclusions/melt_inclusion_datasets.xlsx`"
6. Jupyter notebook for describing how starting compositions were chosen from the full datasets: "`Melt_Inclusions/Choosing_MI_Kilauea.ipynb`"

2 Abbreviations

All models are given by name and full citation in the first appearance in the text and thereafter referred to by a model name or abbreviation only: "Allison" [**Allison2019**], "Eguchi" [Eguchi and Dasgupta 2018], "Frost" [Frost 1991], "Gaillard" [Gaillard et al. 2003], "Iacono-Marziano" or "IM" [Iacono-Marziano et al. 2012], "Kress and Carmichael" or "KC" [Kress and Carmichael 1991], "Liu" [Liu et al. 2005], "Moore" [Moore et al. 1998], "Muth" [Muth and Wallace 2021], "Nash" [Nash et al. 2019], "Shishkina" [Shishkina et al. 2014], and "VolatileCalc" or "VC" [referring to both the original model equations and the Excel VisualBasic app: Dixon et al. 1995; Newman and Lowenstern 2002].

3 Model Options

All model options chosen for the calculations in this manuscript are listed in Table 1. Equilibrium constants are for the following reactions:

K_{H_2O}: $H_2O = H_2 + 0.5O_2$
K_{CO_2}: $CO + 0.5O_2 = CO_2$
K_{H_2S-1}: $0.5S_2 + H_2O = H_2S + 0.5O_2$
K_{H_2S-2}: $H_2S + 1.5O_2 = SO_2 + H_2O$
K_{SO_2}: $0.5S_2 + O_2 = SO_2$
K_{CH_4}: $CO_2 + 2H_2O = CH_4 + 2O_2$
K_{OCS}: $3CO + SO_2 = 2CO_2 + OCS$

Modeling tool settings used to generate our results are included in the Supplement in the folder "`Model_Outputs/resolved_inputs`". Note that for Sulfur_X, the user can define a value for the B constant term to tweak the relationship between Fe and S redox (e.g., to be consistent with XANES data for the system as performed here for Kīlauea). Because of a paucity of experimental data for C-O-H-S-bearing systems at $P < \sim 250$ bars, instability in the computed combined K_D can trend to infinity in special or unusual cases [Ding et al. 2023]. To mitigate this, Sulfur_X allows the user to specify an arbitrary increase in the combined K_D value at each pressure for the lowest pressure steps. In our model runs, we iteratively adjusted this behavior as well as other computational settings that control solver convergence until obtaining a Sulfur_X run with no or minimal low-pressure instability.

Table 1: Model parameterizations used by each tool. See notes and reference key below the table.

Parameter	VolFe	Sulfur_X	D-Compress	MAGEC	EVo
SOLUBILITY					
H_2O	H24 FigS2*	IM12 Eq11[1]*	IM12 Eq11[1]*	IM12 Eq11[2]*	B15 Table1-basalt*
CO_2	$^{\xi,\zeta,\theta}$D95 $^{\psi}$TH94*	IM12 Eq7[1]*	IM12 Eq7[1]*	IM12 Eq7[1]*	B15 Table1-basalt*
S^{2-}	O21 Eq10.34[3]*	D23 Table3-RxnI	N/A	N99 Eq3*	O21 Eq10.34*
S^{6+}	OM22 Eq12a[3]*	D23 Table3-RxnII	N/A	N/A	N/A
SO_2	N/A	N/A	B15 Table1-basalt*	N/A	N/A
H_2S	H24 FigS6-basalt*	N/A	B15 Table1-basalt*	N/A	N/A
H_2	H24 TableS4-basalt*	N/A	B15 Table1-basalt*	G03 Sec5.4	G03 Sec5.4*
CH_4	A13 Eq7a	N/A	0*	A13 Eq7a	A13 Eq7a
CO	H24 TableS4	N/A	0*	A15 Eq10	A15 Eq10
O_2	N/A	N/A	0*	N/A	N/A
S_2	N/A	N/A	0*	N/A	N/A
MELT SPECIATION					
$Fe^{3+}/\Sigma Fe$	KC91 EqA5, A6	KC91 Eq7	KC91 Eq7[4]	SY24 Eq6-9, Table2*	KC91 Eq7
$S^{6+}/\Sigma S$	N/A*	$^{\xi,\psi}$MW21 $^{\zeta}$MW21[5] $^{\theta}$OM22 Eq13*	N/A	SY24 Eq10*	N19 Eq11*
FUGACITY COEFFICIENT					
H_2O	HP91 Eq4, 6, A1-3, Table1*	SS92*	HP91 EqA2-3	PR76 Eq15	HP91
CO_2	SS92*	N/A	SS92	SS92	HP91
SO_2	SS92[6]*	SS92	SS92	SS92	SS92
H_2S	SS92[7]*	SS92	SS92	SS92	SS92
O_2	SS92*	N/A	SS92	SS92	SS92
H_2	SW64 Eq4*	N/A	SW64 Eq4	SS92	SW64 Eq4
CO	SS92*	N/A	SS92	SS92	SS92
CH_4	SS92*	N/A	SS92	SS92	SS92
S_2	SS92*	N/A	SS92	SS92	SS92
OCS	SS92*	N/A	N/A	SS92	N/A

continued on next page

Tablele 1, continued.

Parameter	VolFe	Sulfur_X	D-Compress	MAGEC	EVo
EQUILIBRIUM CONSTANTS					
K_{H_2O}	OK77 Table1-d	N/A	OK77 Table1-d	SL22[8]	EVo RTD[9]
K_{CO_2}	OK77 Table1-c	N/A	OK77 Table1-c	SL22[8]	EVo RTD[9]
$K_{H_2S\text{-}1}$	OK77 Table1-h*	N/A	OK77 Table1-h	SL22[8]	EVo RTD[9]
$K_{H_2S\text{-}2}$	N/A	D23 Eq4	N/A	N/A	N/A
K_{SO_2}	OK77 Table1-f*	N/A	OK77 Table1-f	SL22[8]	EVo RTD[9]
K_{CH_4}	OK77 Table1-e*	N/A	OK77 Table1-e	SL22[8]	EVo RTD[9]
K_{OCS}	M19 Eq8	N/A	N/A	SY24[8]	N/A
OXYGEN FUGACITY					
NNO	F91 Table1	N/A	F91 Table1	H08 Table7[10]	F91 Table1
FMQ	F91 Table1-FMβQ*	F91 Table1-FMβQ	F91 Table1-FMβQ[11]	H08 Table7[12]	F91 Table1-FMβQ
Redox input options	$Fe^{3+}/\Sigma Fe$ $FeO + Fe_2O_3$ ΔFMQ ΔNNO $\log_{10} fO_2$ $S^{6+}/\Sigma S$	ΔFMQ	ΔNNO	$\log_{10} fO_2$ $Fe^{3+}/\Sigma Fe$ $S^{6+}/\Sigma S$ ΔIW ΔFMQ ΔNNO	$Fe^{3+}/\Sigma Fe$ $FeO + Fe_2O_3$ ΔFMQ ΔNNO fO_2 (bar)

Notes: Parameterisations used for all calculations, except where specified for individual melt compositions: $^{\xi}$MORB, $^{\zeta}$Kīlauea, $^{\theta}$Fuego, and $^{\psi}$Fogo. * Indicates other parameterizations are available within the tool. [1]Hydrous parameters; [2]Hydrous web-app parameters; [3]Including H_2O dilution; [4]At 1 bar; [5]Using 6.5 as the last constant value; [6]As modified in Fig. S1 of [Hughes et al. 2023]; [7]As modified in Fig. S1 of [Hughes et al. 2024b]; [8]Based on NIST-JANAF tables [Chase 1996]; [9]Based on JANAF tables [Chase et al. 1998]; [10]References of Frost [1991] and O'Neill and Pownceby [1993]; [11]For this study only because not included in the original tool; [12]References of O'Neill [1987].

References: A13 [Ardia et al. 2013]; A15 [Armstrong et al. 2015]; B15 [Burgisser et al. 2015]; D95 [Dixon et al. 1995]; D23 [Ding et al. 2023]; EVo RTD (`https://evo-outgas.readthedocs.io`); F91 [Frost 1991]; G03 [Gaillard et al. 2003]; H08 [Hirschmann et al. 2008]; HP91 [Holland and Powell 1991]; H24 [Hughes et al. 2024b]; IM12 [Iacono-Marziano et al. 2012]; KC91 [Kress and Carmichael 1991]; M19 [Moussallam et al. 2019]; MW21 [Muth and Wallace 2021]; N99 [Nzotta et al. 1999]; N19 [Nash et al. 2019]; OK77 [Ohmoto and Kerrick 1977]; O21 [O'Neill 2021a]; OM22 [O'Neill and Mavrogenes 2022]; PR76 [Peng and Robinson 1976]; SW64 [Shaw and Wones 1964]; SS92 [Shi and Saxena 1992]; SL22 [Sun and Lee 2022]; SY24 [Sun and Yao 2024]; TH94 [Thibault and Holloway 1994].

4 How to cite models

When using model results in a publication, the specific model used should be: 1. well justified (see main text section ??; and 2. properly cited. The version of the code should be included; model options used within the tool should be clearly recorded and cited appropriately (including if implemented via another tool); and a notebook or script to recreate the calculations and all model results should be included as supplementary material. Here we provide an example suitable for a publication:

> Closed- and open-system degassing curves were modeled using VolFe v1.0 [Hughes et al. 2025] using solubility functions from Dixon et al. [1995], Ardia et al. [2013], O'Neill [2021b], O'Neill and Mavrogenes [2022], and Hughes et al. [2024a], fugacity coefficients from Shaw and Wones [1964], Holland and Powell [1991], Shi and Saxena [1992], and Hughes et al. [2023, 2024b], homogeneous vapor equilibria constants from Ohmoto and Kerrick [1977] and Moussallam et al. [2019], Fe^{3+}/Fe_T from Kress and Carmichael [1991], FMQ and NNO buffers from Frost [1991], and sulfide and anhydrite saturation from Chowdhury and Dasgupta [2019], O'Neill [2021b], and Wieser and Gleeson [2023]. Specific equations used are detailed in Table 4. Parameterizations for model dependent variables were chosen where the calibration dataset most closely matched the composition of the MI of interest. Test calculations show that this selection of model dependent variables in VolFe is able to reproduce saturation pressures from the experiments of "X" to within 20%, which are close in composition to our MI. A full list of model configuration options and notebooks for reproducing test calculations and modeling outputs presented here are in the Supplementary Material.

5 Vapor

The vapor composition during degassing is quite different between different tools (Figure 1). Here we expand on potential differences in the tools that may be contributing to these changes.

5.1 Homogeneous vapor equilibrium constants

The equilibrium constants for homogeneous equilibria in the vapor (K, which are independent of P and melt composition but depend on T) control the vapor composition at a given set of conditions and therefore differences in K can cause variations between model outputs. These K can be calculated from model results using the vapor composition (x_i^v and f_{O_2}) at 1 bar when $\gamma_i = 1$ (i.e., $f_i = x_i^v P$ at 1 bar) according to:

$$H_2(v) + 0.5O_2(v) = H_2O(v) \tag{1}$$

$$K_H = \frac{f_{H_2O}}{f_{H_2} f_{O_2}^{0.5}} = \frac{\gamma_{H_2O} x_{H_2O}^v P}{\gamma_{H_2} x_{H_2}^v P f_{O_2}^{0.5}} \tag{2}$$

$$CO(v) + 0.5O_2(v) = CO_2(v) \tag{3}$$

$$K_C = \frac{f_{CO_2}}{f_{CO} f_{O_2}^{0.5}} = \frac{\gamma_{CO_2} x_{CO_2}^v P}{\gamma_{CO} x_{CO}^v P f_{O_2}^{0.5}} \tag{4}$$

$$0.5S_2(v) + O_2(v) = SO_2(v) \tag{5}$$

$$K_S = \frac{f_{SO_2}}{f_{S_2}^{0.5} f_{O_2}} = \frac{\gamma_{SO_2} x_{SO_2}^v P}{\left(\gamma_{S_2} x_{S_2}^v P\right)^{0.5} f_{O_2}} \tag{6}$$

$$CH_4(v) + 0.5O_2(v) = CO(v) + 2H_2(v) \tag{7}$$

$$K_{CH} = \frac{f_{CO} f_{H_2}^2}{f_{CH_4} f_{O_2}^{0.5}} = \frac{\gamma_{CO} x_{CO}^v P \left(\gamma_{H_2} x_{H_2}^v P\right)^2}{\gamma_{CH_4} x_{CH_4}^v P f_{O_2}^{0.5}} \tag{8}$$

$$SO_2(v) + H_2(v) = H_2S(v) + O_2(v) \tag{9}$$

$$K_{HS} = \frac{f_{H_2S} f_{O_2}}{f_{SO_2} f_{H_2}} = \frac{\gamma_{H_2S} x_{H_2S}^v P f_{O_2}}{\gamma_{SO_2} x_{SO_2}^v P \gamma_{H_2O} x_{H_2}^v P} \tag{10}$$

Table 2: Parameterizations used for model dependent variables in degassing calculations using VolFe.

Variable	Reference
Fugacity coefficients	
O_2	Shi and Saxena [1992]
CO	Shi and Saxena [1992]
H_2	Shaw and Wones [1964]
S_2	Shi and Saxena [1992]
CO_2	Shi and Saxena [1992]
H_2O	Eq. (4, 6: T > 673 K, A1-3) & Table 1 in Holland and Powell [1991]
SO_2	Shi and Saxena [1992] as modified in Fig. S1 of Hughes et al. [2023]
CH_4	Shi and Saxena [1992]
H_2S	Shi and Saxena [1992] as modified in Fig. S1 of Hughes et al. [2024b]
OCS	Shi and Saxena [1992]
Homogeneous vapor equilibrium constants	
CO_2	Reaction (c) in Table 1 of Ohmoto and Kerrick [1977]
H_2O	Reaction (d) in Table 1 of Ohmoto and Kerrick [1977]
SO_2	Reaction (f) in Table 1 of Ohmoto and Kerrick [1977]
CH_4	Reaction (e) in Table 1 of Ohmoto and Kerrick [1977]
H_2S	Reaction (h) in Table 1 of Ohmoto and Kerrick [1977]
OCS	Eq. (8) in Moussallam et al. [2019]
Solubility functions	
H_2O_T	Fig. S2 in Hughes et al. [2024b]
$CO_{2,T}$	Bullet (5) in Dixon et al. [1995]
$H_{2,mol}$	Basalt from Table S4 in Hughes et al. [2024b]
CO_{mol}	Table S4 in Hughes et al. [2024b]
$CH_{4,mol}$	Eq. (7a) from Ardia et al. [2013]
*S^{2-}	Eq. (10.43) from O'Neill [2021a] (including H_2O dilution)
SO_4^{2-}	Eq. (12a) from O'Neill and Mavrogenes [2022] (including H_2O dilution)
H_2S_{mol}	Basalt from Fig. S6 in Hughes et al. [2024b]
fO_2 related parameters	
Fe^{3+}/Fe_T	Eq. (A-5, 6) from Kress and Carmichael [1991] (no H_2O dilution)
FMQ	FMβQ in Table 1 of Frost [1991]
NNO	NiNiO in Table 1 of Frost [1991]
Other	
S^{2-}CSS	Eq. (10.34, 10.43, 10.45, 10.46) in O'Neill [2021a]
S^{6+}CAS	Eq. (8) in Chowdhury and Dasgupta [2019] using PySulfSat v1.0.2 [Wieser and Gleeson 2023]

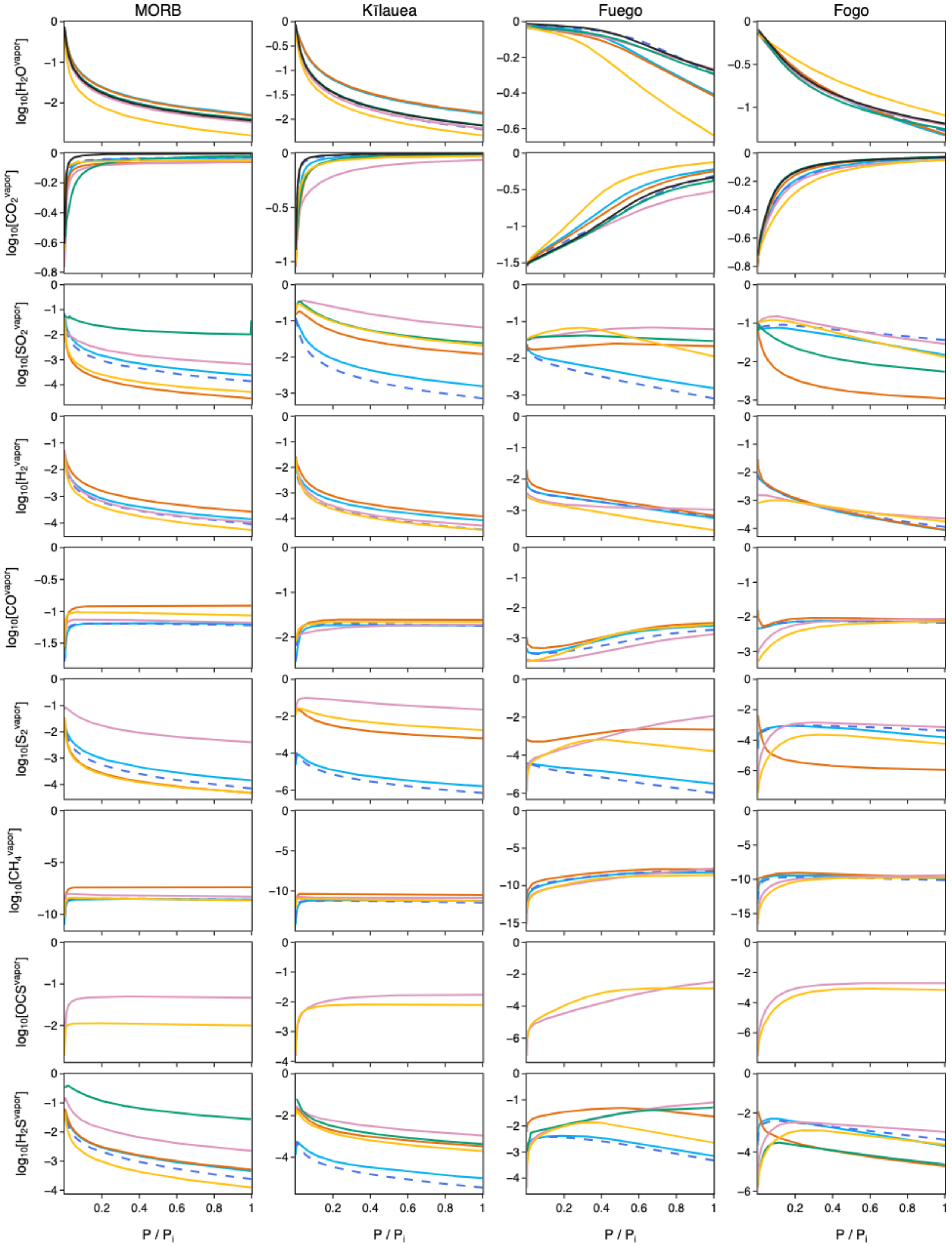

Figure 1: Vapor composition against normalized P for each system using the different tools.

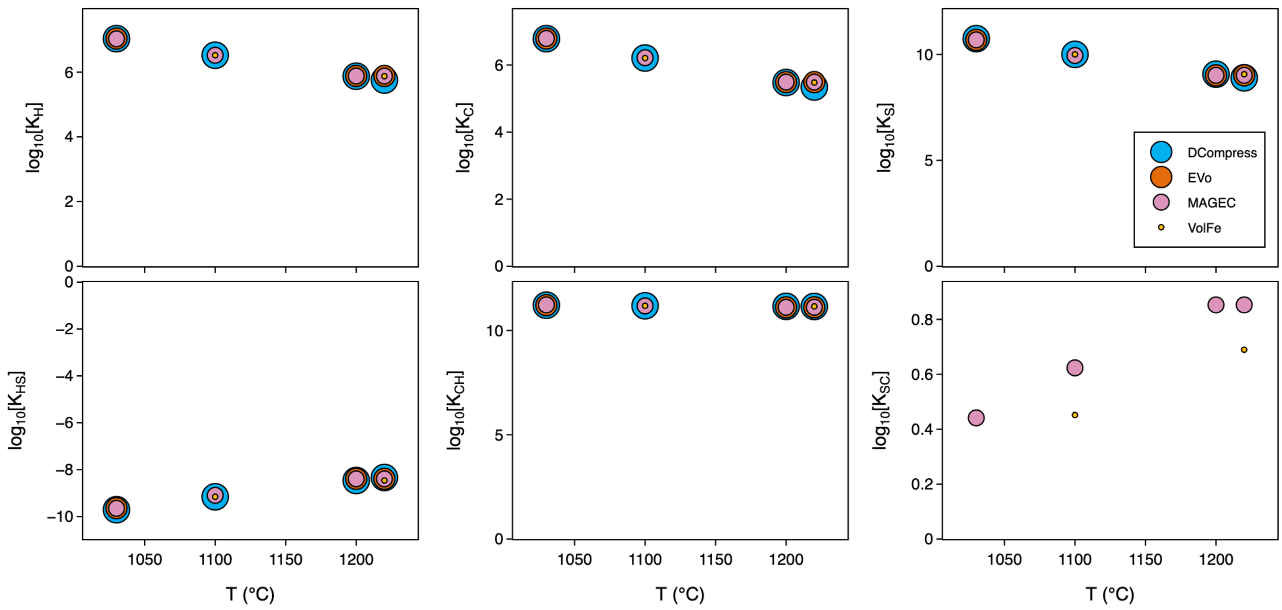

Figure 2: Equilibrium constants for each tool and system (i.e., different temperatures). Calculated at P = 1 bar as described in the text.

$$OCS(v) = CO(v) + 0.5S_2(v) \tag{11}$$

$$K_{SC} = \frac{f_{CO} f_{S_2}^{0.5}}{f_{OCS}} = \frac{\gamma_{CO} x^v_{CO} P \left(\gamma_{S_2} x^v_{S_2} P\right)^{0.5}}{\gamma_{OCS} x^v_{OCS} P} \tag{12}$$

Although these specific reactions and K may not have been used in the tools directly, these are six independent reactions that describe a vapor containing O_2, H_2, CO, S_2, H_2O, CO_2, SO_2, CH_4, H_2S, and OCS (i.e., no reaction can be made from a combination of the other reactions). Hence, this represents a minimum number of K in the system. Equilibrium constants are only calculated if the tool outputs all required x_i^v for a specific reaction (e.g., Sulfur_X, D-Compress, and EVo do not include x^v_{OCS} and therefore K_{OCS} cannot be calculated) and if the system calculation reached $P \leq 1$ bar (e.g., VolFe calculations for Fuego and MORB did not reach P = 1 bar and therefore values are not calculated; Figure 2). D-Compress and VolFe use Ohmoto and Kerrick [1977], whereas Evo and MAGEC use in-house versions based on NIST/JANAF databases. Implications of these variations are discussed in the main text.

5.2 Fugacity coefficients

We can also use the K_i equations to investigate the difference between γ_i in the different tools, which only depend on P and T (i.e., not melt composition). Individual γ_i cannot be calculated, but ratios of various combinations can be. If the ratios are similar, this suggests the tools are using similar γ_i (although the γ_i could be perfectly off-setting each other, but that would be unlikely). As for calculating K_i, the ratios of γ_i are only calculated if all the necessary vapor species are calculated and the calculation reaches ≤ 1 bar as K_i must also be calculated.

Most tools are using very similar values for γ_{H_2}, γ_{S_2}, γ_{SO_2}, γ_{H_2S}, and γ_{OCS}, as can be seen by comparing $\gamma_{S_2}^{0.5}/\gamma_{SO_2}$, $[\gamma_{SO_2}\gamma_{H_2}]/\gamma_{H_2S}$, and $\gamma_{OCS}/[\gamma_{CO}\gamma_{S_2}^{0.5}]$ in Figure 3. This makes sense given all tools are using Shi and Saxena [1992] for γ_{S_2}, γ_{SO_2}, γ_{H_2S}, and γ_{OCS}. The differences for VolFe at low-P for γ_{SO_2} and γ_{H_2S} are due to modifications described in Hughes et al. [2023, 2024b] and are therefore expected. Most tools use Shaw and Wones [1964] for γ_{H_2}, except MAGEC that uses a modified version of Shi and Saxena [1992], and therefore the similarity in γ_{H_2} is good. Most tools use equations from Holland and Powell [1991] (although not the same specific equations), except MAGEC that uses eq. (15) from Peng and Robinson [1976], which are likely giving different values leading to the differences seen in $\gamma_{H_2}/\gamma_{H_2O}$ (Figure 3). Values for γ_{CO_2}, γ_{CO}, and γ_{CH_4} seem similar between tools (e.g., $\gamma_{CH_4}/[\gamma_{CO}(\gamma_{H_2})^2]$ and $\gamma_{CH_4}/[\gamma_{CO_2}(\gamma_{H_2})^2]$) in Figure 3). All tools use Shi and Saxena [1992] for γ_{CH_4} and γ_{CO}, and most also use this for γ_{CO_2}, except EVo that uses Holland and Powell [1991]. There appear to be some minor variations in $\gamma_{CO}/\gamma_{CO_2}$, suggesting minor differences in γ_{CO} and/or γ_{CO_2}.

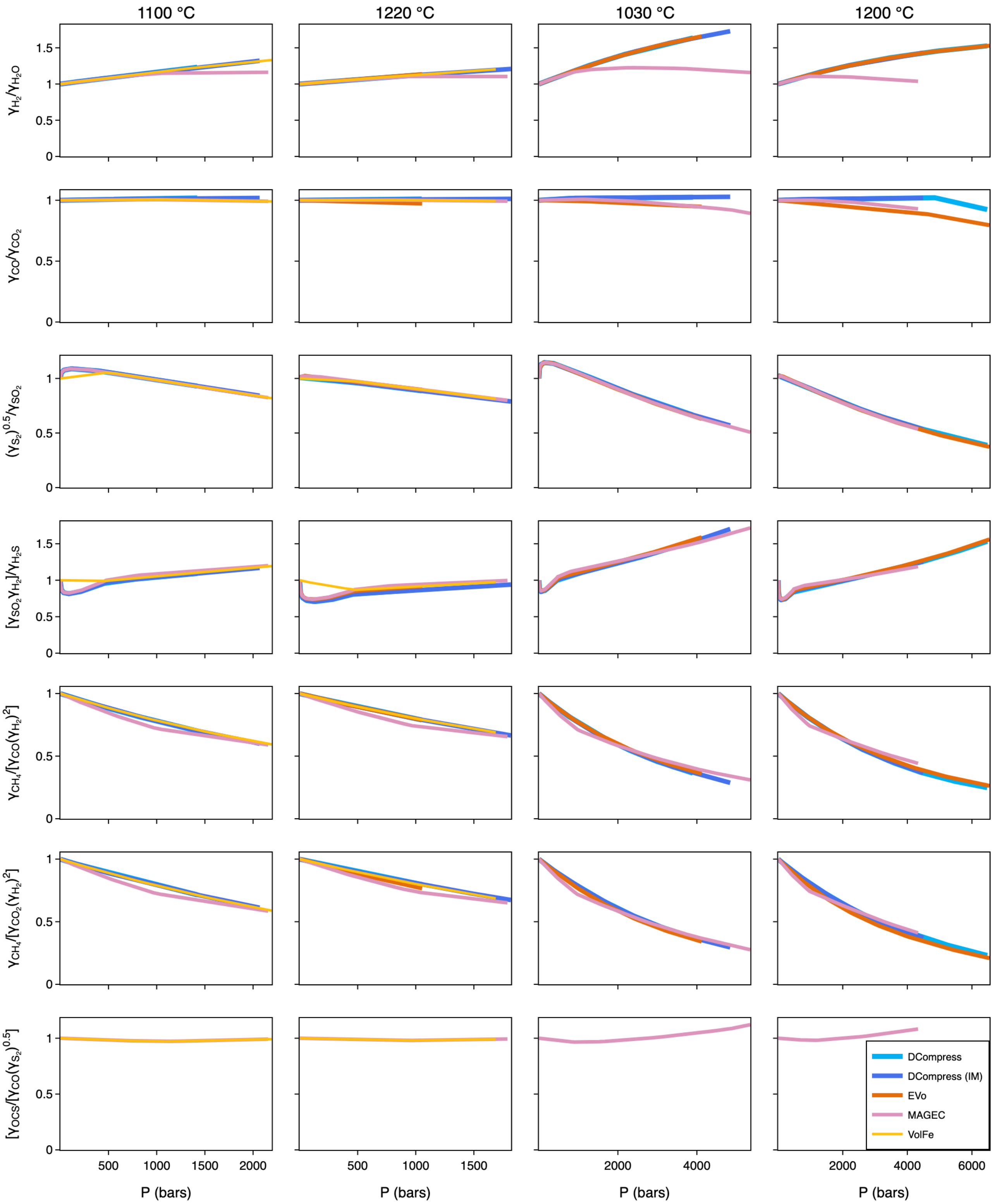

Figure 3: Ratios of fugacity coefficients (γ_i calculated using equilibrium constant equations in the text)

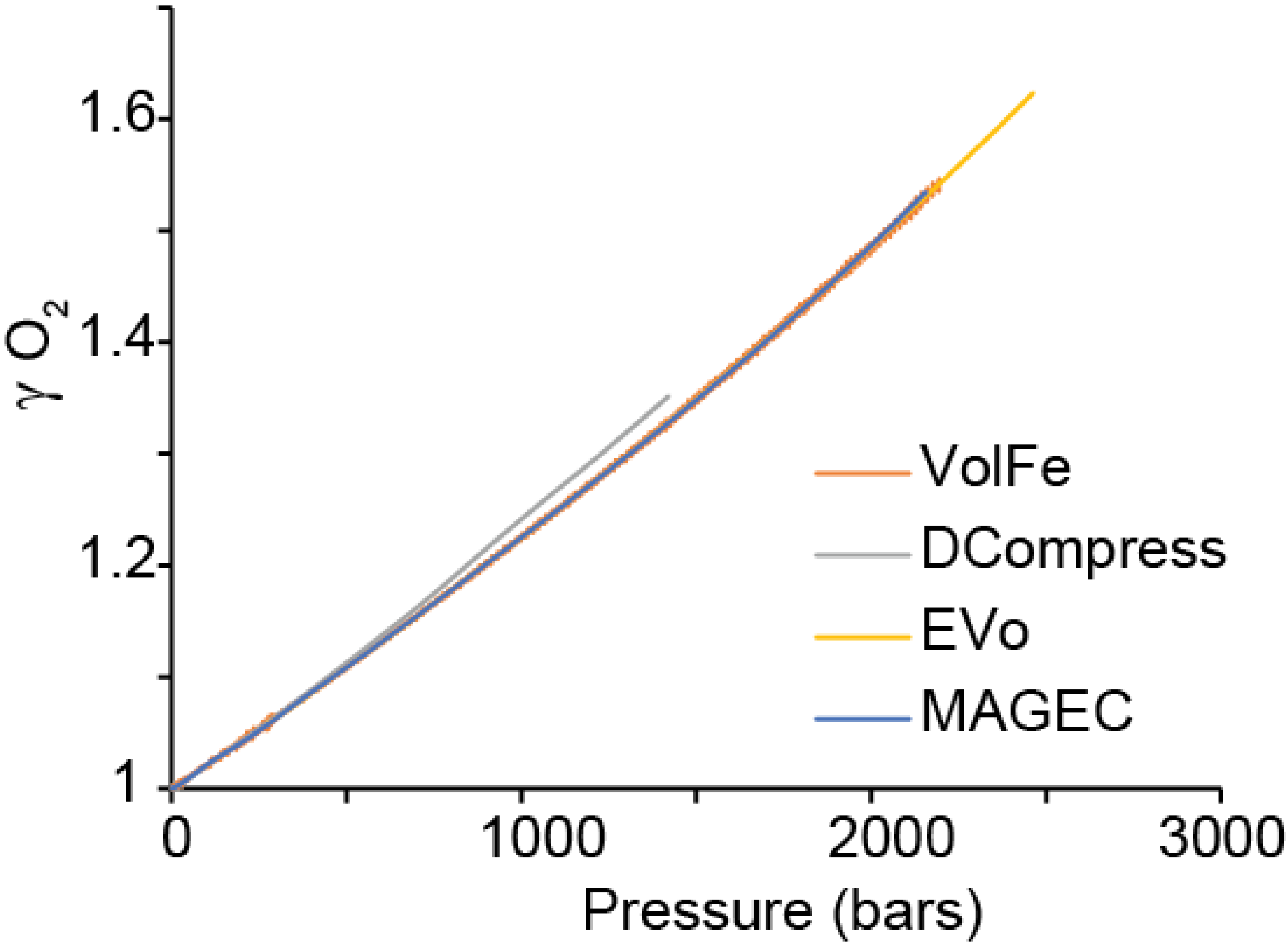

Figure 4: Non-ideal factor γ for O_2 vs. pressure for four models.

O_2 in the vapor is highly non-ideal, and the deviation from ideality, the fugacity coefficient (γ), is dependent upon P and T. We can test whether models are treating the non-ideality of O_2 in the same way by calculating γ_{O_2} from the definition of fugacity as a function of P:

$$fO_2 = \gamma_{O_2} \times x^v_{O_2} \times P. \tag{13}$$

γ_{O_2} is unity for an ideal gas, but deviates strongly for a real gas, particularly at high P [e.g., Shi and Saxena 1992]. Supplementary figure 4 shows nearly perfect overlap, with γ_{O_2} from unity at 1 bar up to ~1.5 at 2,000 bars, meaning that this is not a factor in the divergence of computed redox variables and is expected given all tools use the model of Shi and Saxena [1992] for γ_{O_2}. Hence, differences in γ_{O_2} are not driving differences between model outputs.

5.3 Molar mass balance in the vapor

One difficulty when using models from the literature is that some post-processing calculations cannot be carried out because they are at odds with either model assumptions or numerical limitations. The molar fraction of O_2 in the vapor, $x^v_{O_2}$, is a prime example of such limitations. Sulfur_X does not include O_2 in the vapor calculation and therefore is not included in the following discussion. All other models assume that the sum of the molar fractions of all species in the vapor is equal to one, and thus one should be able to deduce the fraction of one species by calculating one minus the sum of all the others species. $x^v_{O_2}$ is notoriously small (typically 10^{-15}–10^{-11}), whereas $x^v_{H_2O}$ is typically 10^{-1}. This means that numerical limitations may be encountered because most models report (and calculate) in double precision (~15 significant digits). Figure 5 shows $x^v_{O_2}$ calculated by difference versus $x^v_{O_2}$ values output by the models. The bottom row shows the residual of the closure calculation $\Sigma x^v_i - 1$ (the sum of all vapor species minus 1) at each P using Kīlauea as a representative example.

D-Compress uses Turbo Delphi 2006 with TuboPower SysTools BCD allowing for floating point precision to 26 significant digits. All other models use Python or MATLAB's default double precision (IEEE 754 binary64) which is precise to 15–17 significant digits. This matters when very small values are converted between decimal representation (as we read it) and binary (as the computer reads it). Our plotted $x^v_{O_2}$ by difference and residual values were likewise computed in binary64

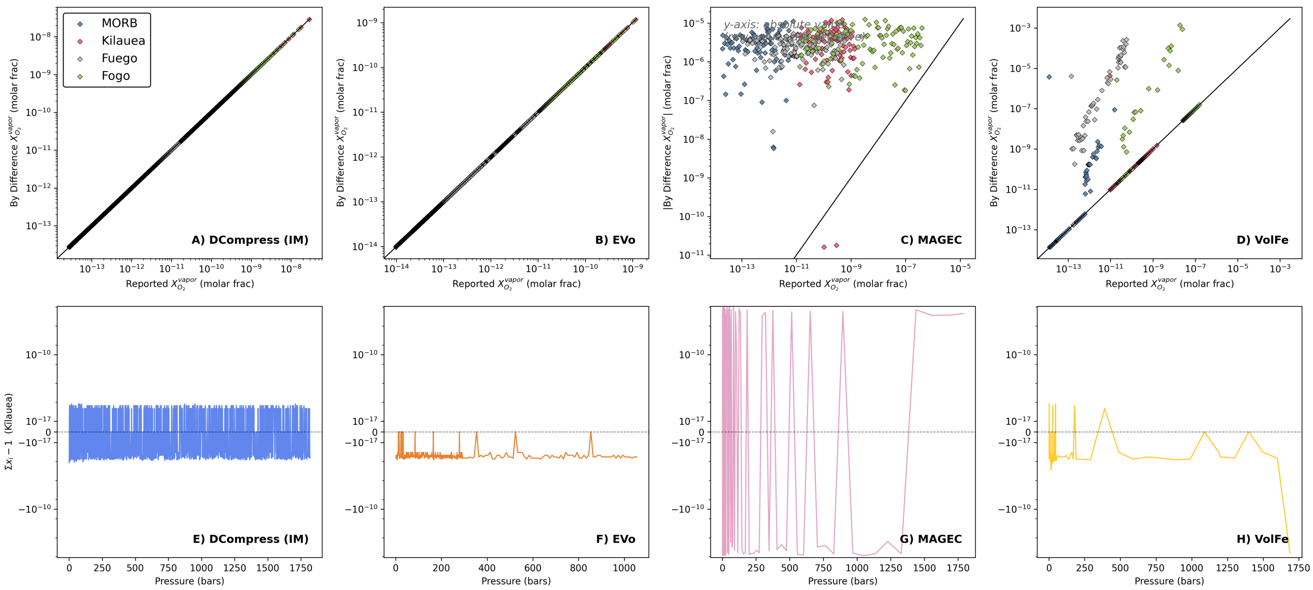

Figure 5: Molar fraction of O_2 in the vapor as reported by modeling tools vs. calculated by difference for all models that report it: D-Compress, EVo, MAGEC, and VolFe. Some by difference values computed for MAGEC were negative and so we plot the absolute values in order to plot on a logarithmic axis. The bottom row shows the residual of the closure calculation (the departure from zero of the sum of all vapor species minus 1) for Kīlauea runs.

from each model's CSV output, meaning that residuals below this floor cannot be resolved regardless of how the values were computed internally. In other words, any trailing digits are meaningless noise from rounding error.

D-Compress closure shows symmetric noise at the $\sim 10^{-17}$ level scattered around zero, reflecting only float64 rounding caused by our by-difference calculation and indicating higher precision within the model itself. The closure plot in EVo reveals systematically negative residuals $\sim 10^{-16}$–10^{-17}, indicating a normalization step within the solver capping sums to one. At reducing conditions below $\log fO_2$ of -15 ($\sim \Delta$IW-3), $x^v_{O_2}$ cannot be recovered by-difference. Data by difference from MAGEC oscillate between negative and positive and, for that reason, we plot the absolute value in the 1:1 figure for MAGEC. The MAGEC output files used in this comparison were exported to 5 significant figures, resulting in the closure error of $\sim 10^{-5}$–10^{-6}. Each species is rounded independently at output time, so per-P step sign distributions are random and centered on zero (output-format rounding, not algorithmic bias). This does not indicate anything wrong with MAGEC's internal solver, but this output precision means that fO_2 cannot be reliably recovered by difference from the exported MAGEC outputs used here. Most VolFe outputs reflect the same expected double-precision behavior as EVo but show closure failure in certain cases, specifically for high-CO_2, low-H_2O points at the mid-P. This is likely the result of its iterative Newton-Raphson solver converging on a near-solution and exiting before normalization of gas species to one can be enforced. Adjusting the solver tolerance dependent upon composition could improve the recoverability of $x^v_{O_2}$ in certain cases.

High precision is not typically required to answer geologic questions, but the impact can be significant due to repeated computation within the code itself before a result gets returned to a user. Likewise, if precision is truncated when the code writes results to disk, any calculation those values are fed back into will never be better than the number of truncated digits, and this would almost certainly go undetected, since floating point error thereafter will mask the previous truncation. It is also worth flagging floating point rounding behavior in Microsoft Excel. Although computations within Excel adhere to IEEE 754 binary64, when any value is *displayed* in the user interface, it is truncated to 15 significant figures, replacing the final two with zeroes. Additionally, if a file with numbers exceeding 11 digits is *saved* from Excel, it's "General" format defaults to scientific notation and may silently drop several significant figures from all values [Griffiths 2022]. This effect has been noted as particularly damaging for data analysis in the genome biology literature [Ziemann et al. 2016].

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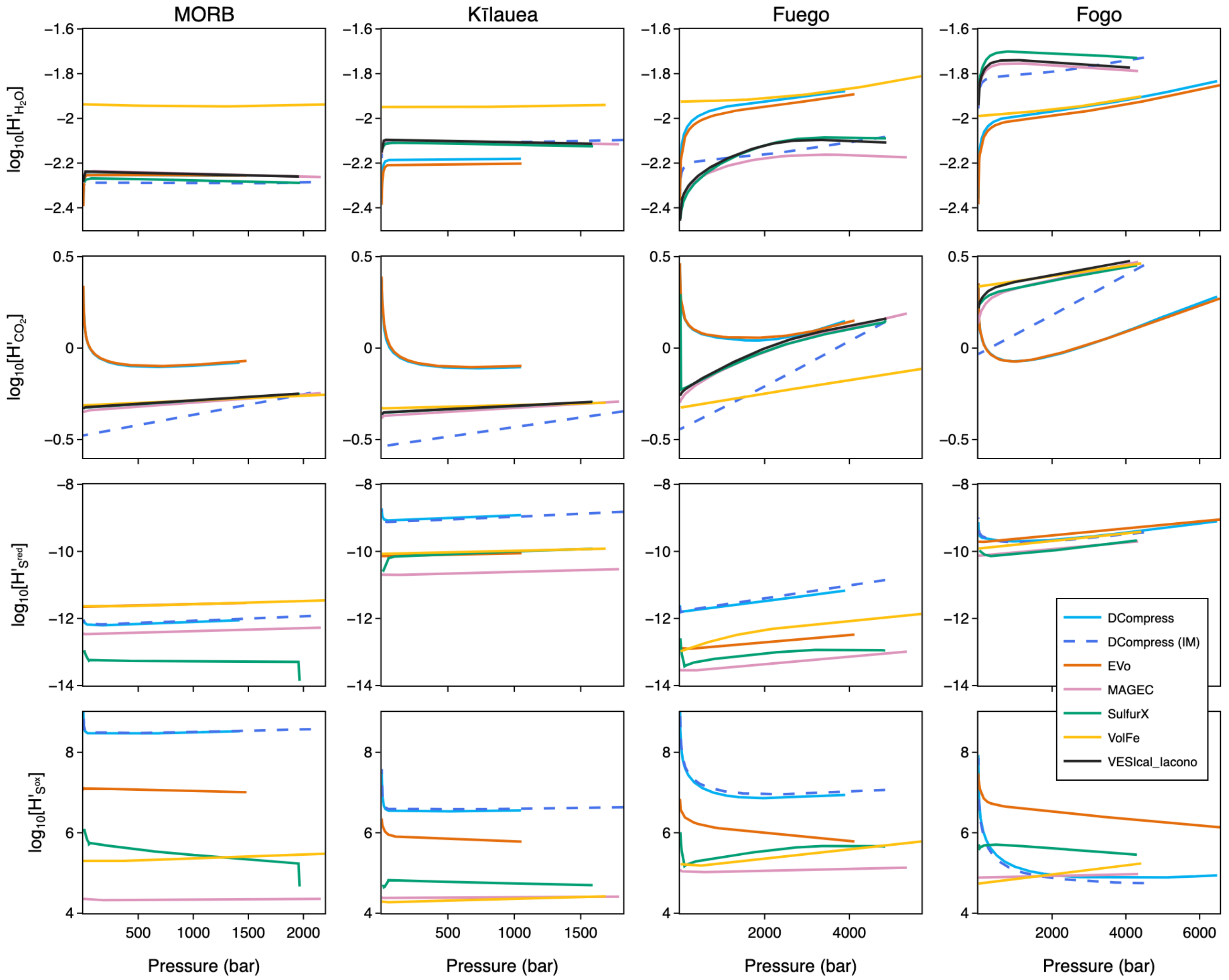

Figure 6: Volatile proportionality factors (eq. 15 with all tools for all basalts showing the change in H_2O, CO_2, reduced S (S^{red}), and oxidized S (S^{ox}) proportionality factors (H_i') as a function of pressure during degassing.

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